Problem Statement: Power System Fault Detection and Classification using Machine Learning

Power distribution systems are critical for delivering uninterrupted electricity, but they are vulnerable to faults such as Transformer Failure, Line Breakage, and Overheating. Delayed or inaccurate fault identification can lead to equipment damage, power outages, increased maintenance costs, and reduced grid reliability.

Traditional fault detection methods often require manual analysis and may not provide rapid diagnosis. Therefore, there is a need for an intelligent system that can automatically analyze electrical parameters and accurately classify different types of power system faults.

This project aims to develop PowerGuard AI, a Machine Learning-based fault detection and classification system that uses electrical measurements such as voltage, current, power load, temperature, wind speed, weather conditions, maintenance status, component health, fault duration, and downtime to predict the type of fault. The system provides a user-friendly web interface for prediction and an analytics dashboard for data visualization, supporting faster fault diagnosis and improved maintenance planning.